Chapter 9, Problem 2P

Problem

A current source in a linear circuit has

$$i_s = 15\cos(25\pi t + 25°)\text{ A}$$

(a) What is the amplitude of the current?

(b) What is the angular frequency?

(c) Find the frequency of the current.

(d) Calculate *is* at *t* = 2 ms.

Step-by-step solution

Step 1 of 4

Consider a current source i_s in a linear circuit is $15\cos(25\pi t + 25°)\text{ A}$.

(a)

Write the general form of time domain expression for current, $i(t)$.

$$i(t) = I_m\cos(\omega t + \theta°)\text{ A}$$

Compare the general form of time domain expression for current and the current source, i_s.

The amplitude of the current is,

$$I_m = 15\text{ A}$$

Therefore, the Amplitude is $\boxed{15\text{ A}}$.

Step 2 of 4

(b)

Calculate the value of angular frequency, ω.

Write the expression for angular frequency, ω.

$$\begin{aligned}\omega &= 25\pi\\ &= 25(3.14)\\ &= 78.54\text{ rads/s}\end{aligned}$$

Hence, the angular frequency ω is $\boxed{78.54\text{ rad/s}}$.

Step 3 of 4

(c)

Calculate the frequency f.

Write the expression for the frequency f.

$$\begin{aligned} f &= \frac{\omega}{2\pi} \\ &= \frac{25\pi}{2\pi} \\ &= 12.5\ \text{Hz} \end{aligned}$$

Hence, the frequency f is $\boxed{12.5\ \text{Hz}}$.

Step 4 of 4

(d)

Calculate i_s at $t = 2\ \text{ms}$.

Substitute $t = 2\ \text{ms}$ in $i_s = 15\cos(25\pi t + 25°)\ \text{A}$.

Convert $25°$ into radians.

$$\begin{aligned} \theta &= 25° \\ &= (25°)\left(\frac{\pi}{180°}\right) \\ &= 0.436\ \text{rads} \end{aligned}$$

Calculate the value of i_s at $t = 2\ \text{ms}$.

$$\begin{aligned} i_s &= 15\cos(25\pi t + 0.436) \\ &= 15\cos\left(\left(25 \times \pi \times 2 \times 10^{-3}\right) + 0.436\right) \\ &= 15\cos(0.5931) \\ &= 15(0.8292) \end{aligned}$$

$$= 12.44\ \text{A}$$

Hence, the value of current i_s at $t = 2\ \text{ms}$ is $\boxed{12.44\ \text{A}}$.